

Computer Networking — Complete Comprehensive Guide & MCQ Bank

Accenture & Campus Placement Drive Revision | Theory, Mnemonics, Diagrams, Tables & Answer Explanations

1. OSI Model ★★★★★

This is probably the most important networking topic for MCQs.

Definition: OSI (Open Systems Interconnection) model is a conceptual framework that divides network communication into 7 layers.

7 → Application 6 → Presentation 5 → Session 4 → Transport 3 → Network 2 → Data Link 1 → Physical

Easy mnemonic (From Layer 7 → 1):

All People Seem To Need Data Processing

Application | Presentation | Session | Transport | Network | Data Link | Physical

Layer 1 — Physical

What does it do? Transmits raw bits over the physical medium (e.g. 101101010100).

It deals with: Electrical signals, Radio signals, Cables, Connectors, Voltage, Physical transmission.

Examples: Ethernet cable, Fiber optic cable, Radio waves, Hubs/repeaters.

Data unit: Bits

MCQ: Which OSI layer deals with transmission of raw bits? → **Physical Layer**

Layer 2 — Data Link

Main job: Provides node-to-node communication over a local network and handles framing and MAC addressing.

It deals with: Frames, MAC addresses, Error detection, Switching.

Data unit: Frame

Important protocols/technologies: Ethernet, Wi-Fi (802.11), PPP | **Device:** Switch

MAC Address: A network interface has a MAC address used for local network communication. Example: 00:1A:2B:3C:4D:5E

Two sublayers: Data Link is often divided into **LLC** (Logical Link Control) and **MAC** (Media Access Control). *(For Accenture, know the names; deep IEEE details aren't usually necessary.)*

Layer 3 — Network ★★★★★

Main job: Responsible for logical addressing and routing packets between networks.

Key concept: IP addressing + routing

Examples: IPv4, IPv6, ICMP | **Device:** Router | **Data unit:** Packet

Computer → Router → Another Network → Server (The router determines where packets should go.)

MCQ: Which OSI layer is responsible for routing? → **Network Layer**

Layer 4 — Transport ★★★★★

This is extremely important.

Main job: Provides end-to-end communication between applications/processes.

Main protocols: TCP and UDP

- **TCP (Transmission Control Protocol):** Provides Connection-oriented communication, Reliable delivery, Sequencing, Flow control, Congestion control, Retransmission.
- **UDP (User Datagram Protocol):** Provides Connectionless communication, Low overhead, No delivery guarantee, No retransmission mechanism like TCP.

Data unit: Generally TCP → **Segment** | UDP → **Datagram**

Layer 5 — Session

Main job: Establishes, manages, and terminates communication sessions between applications. (*Think: Session = conversation management*)

Client → Session established → Communication → Session terminated

For Accenture, remember the function rather than going deep into session-layer protocol details.

Layer 6 — Presentation

Main job: Responsible for data representation/translation.

Handles: Data format translation, Encryption/decryption, Compression/decompression, Character encoding.

Application data → Presentation → Convert/format data → Network transmission

Easy memory: Presentation = How data is represented

Layer 7 — Application

Main job: Provides network services to applications/users.

Examples: HTTP, HTTPS, FTP, SMTP, DNS

Important: Application layer doesn't mean "The application itself." It means the layer where network services used by applications operate.

🔥 OSI Layer Cheat Sheet

Layer	Name	Main Concept	Data Unit	Example
7	Application	Network services	Data	HTTP, DNS
6	Presentation	Format / encryption / compression	Data	Encoding, TLS
5	Session	Session management	Data	Session control
4	Transport	End-to-end delivery	Segment / Datagram	TCP / UDP
3	Network	IP / routing	Packet	IP, ICMP
2	Data Link	MAC / framing	Frame	Ethernet, Switch
1	Physical	Signals / bits	Bits	Cable, Hub

🔥 Common Accenture OSI Questions

- **Q1:** A router primarily operates at which OSI layer? → **Layer 3 — Network**
- **Q2:** A switch primarily operates at which layer? → **Layer 2 — Data Link**
- **Q3:** TCP operates at which layer? → **Layer 4 — Transport**

- **Q4:** IP operates at which layer? → **Layer 3 — Network**
- **Q5:** MAC addressing belongs primarily to: → **Layer 2 — Data Link**
- **Q6:** HTTP belongs to: → **Layer 7 — Application**

2. TCP/IP Model ★★★★★

The TCP/IP model is the practical networking model underlying the Internet. A commonly taught version has 4 layers: **4 → Application, 3 → Transport, 2 → Internet, 1 → Network Access.**

Compare with OSI

OSI Layer (7 Layers)	TCP/IP Model (4 Layers)
Application, Presentation, Session	Application
Transport	Transport
Network	Internet
Data Link, Physical	Network Access

Important: OSI = 7 layers | TCP/IP = 4 layers in the common four-layer model.

3. TCP vs UDP ★★★★★

This is a must-know Accenture topic.

TCP (Transmission Control Protocol): Connection-oriented and reliable. Before data transfer, TCP establishes a connection via the Three-way handshake (SYN → SYN-ACK → ACK). Provides Reliable delivery, Sequencing, Acknowledgements, Retransmission, Flow control, Congestion control.

UDP (User Datagram Protocol): Connectionless and lightweight. Doesn't establish a TCP-style connection before sending data. Advantages: Faster/low overhead, No connection establishment, Useful where low latency matters. Examples: DNS queries, Streaming, Online gaming, VoIP.

TCP vs UDP Table

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	✓ Yes	✗ No
Ordering	✓ Yes	✗ No
Retransmission	✓ Yes	✗ No
Handshake	✓ Yes	✗ No
Speed / Overhead	Higher overhead	Lower overhead
Flow control	✓ Yes	✗ No
Congestion control	✓ Yes	✗ No
Use case	Web, file transfer, email	DNS, streaming, gaming

Accenture Trap:

- "Fast because it doesn't establish a connection" → **UDP**
- "Reliable and ordered delivery" → **TCP**

4. TCP Three-Way Handshake ★★★★★

You should definitely know this.

Client Server SYN → (Step 1: Client requests connection) ← SYN-ACK (Step 2: Server accepts & responds) ACK → (Step 3: Client confirms) Connection Established → Data Transmission Begins

Memory: SYN → SYN-ACK → ACK

5. IP Address ★★★★★

Definition: An IP address is a logical address used to identify a device/interface on an IP network and enable packet delivery/routing. Example IPv4: 192.168.1.10

IPv4: Uses **32 bits**. Format: 4 octets (192.168.1.10), each octet is 8 bits ($4 \times 8 = 32$ bits). Total possible addresses: 2^{32} approx 4.29 ext{ billion}.

IPv6: Uses **128 bits**. Example: 2001:0db8:85a3::8a2e:0370:7334. Introduced because IPv4 address space is limited. Total space: 2^{128} addresses.

IPv4 vs IPv6

Feature	IPv4	IPv6
Address size	32-bit	128-bit
Format	Decimal (4 octets)	Hexadecimal (8 blocks)
Example	192.168.1.1	2001:db8::1
Address space	Smaller (~4.29B)	Huge (~ 3.4×10^{38})
Broadcast	Supported	No traditional broadcast (Multicast/Anycast)
Address configuration	DHCP / manual	Supports SLAAC / DHCPv6

Accenture must remember: IPv4 → 32 bits | IPv6 → 128 bits

6. Public vs Private IP ★★★★★

Private IP: Used inside private networks (LANs, homes, offices). Common IPv4 private ranges (RFC 1918):

- **Range 1:** 10.0.0.0/8 (10.0.0.0 – 10.255.255.255)
- **Range 2:** 172.16.0.0/12 (172.16.0.0 – 172.31.255.255)
- **Range 3:** 192.168.0.0/16 (192.168.0.0 – 192.168.255.255)

Public IP: Globally routable on the Internet (e.g., 8.8.8.8).

Important: Private IP = Inside private networks | Public IP = Internet-routable communication.

7. NAT (Network Address Translation) ★★★★★

NAT translates addresses between networks, commonly allowing multiple private IPv4 addresses to share public IPv4 addresses for Internet access.

Private Network: (192.168.1.10, 192.168.1.11, 192.168.1.12) → [NAT] → Public IP → Internet

Why NAT? Major reason: Conserve public IPv4 addresses. It also helps hide internal addressing from direct Internet exposure (though NAT itself is not a complete security mechanism).

8. Networking Devices ★★★★★

Know these very well.

- **Hub (Layer 1 — Physical):** Receives a signal and forwards/broadcasts it to all ports without intelligent destination selection.
- **Switch (Layer 2 — Data Link):** Uses MAC addresses. Forwards frames toward the appropriate port based on its MAC address table.
- **Router (Layer 3 — Network):** Uses IP addresses. Connects and routes packets between different networks.
- **Gateway:** Connects networks using different protocols or architectures. The *default gateway* is the router/interface used to reach outside the local network.

🔥 Device Comparison Table

Device	Main Layer	Uses	Main Function
Hub	L1 (Physical)	Signals	Broadcasts incoming signals to all ports
Switch	L2 (Data Link)	MAC	Connects devices in LAN; forwards based on MAC table
Router	L3 (Network)	IP	Routes packets between different IP networks
Gateway	Varies (L3+)	Depends	Connects different networks/protocols

Very common MCQs:

- Which device forwards packets based on IP addresses? → **Router**
- Which device forwards frames based on MAC addresses? → **Switch**
- Which device broadcasts incoming signals to all ports? → **Hub**

9. HTTP / HTTPS ★★★★★

HTTP (Hypertext Transfer Protocol): Port **80**. Used for web communication between clients and servers.

HTTPS (HTTP + TLS): Port **443**. Provides Confidentiality, Integrity, and Server authentication.

MCQ: A browser communicates securely with a web server. Which protocol? → **HTTPS**

10. DNS (Domain Name System) ★★★★★

Definition: Translates human-friendly domain names (e.g. `www.google.com`) into IP addresses (e.g. `142.250.xxx.xxx`).

Common Port: **53** (Uses both UDP and TCP depending on situation). *Basic MCQ: DNS → Port 53 → Name resolution.*

11. DHCP (Dynamic Host Configuration Protocol) ★★★★★

Definition: Automatically provides network configuration to devices (IP address, Subnet mask, Default gateway, DNS server).

DHCP Process — DORA:

- D** → Discover (Client broadcasts: "Is there a DHCP server?")
- O** → Offer (DHCP server offers IP config)
- R** → Request (Client requests offered config)
- A** → Acknowledgment (Server confirms assignment)

12. FTP (File Transfer Protocol) ★★★★★

Used to transfer files between systems. Traditional FTP uses **TCP 21** for control (and a separate data channel). Secure alternative: **SFTP** runs over SSH on **TCP 22**. (*Don't confuse: FTP → 21 | SSH/SFTP → 22*)

13. SMTP (Simple Mail Transfer Protocol) ★★★★★

Used for sending/relaying email. Common ports: **25** (SMTP relay), **587** (Message submission), **465** (SMTP over TLS). Basic MCQ: SMTP → Sending email → Port 25.

14. SSH (Secure Shell) ★★★★★

Used for encrypted secure remote login, remote command execution, and administration. Common port: **22**.

🔥 Common Protocol-Port Table

Protocol	Purpose	Port	Extra High-Value Protocol	Port
FTP	File transfer	21	DHCP Server / Client	67 / 68
SSH	Secure remote login	22	Telnet	23
SMTP	Email sending / relay	25	SNMP	161
DNS	Domain resolution	53	LDAP	389
HTTP	Web	80	RDP	3389
POP3	Email retrieval	110	Important first set to memorize: 21, 22, 25, 53, 80, 110, 143, 443	
IMAP	Email retrieval / sync	143		
HTTPS	Secure web	443		

🌐 Accenture Computer Networking — Tier 1 MCQ Practice & Complete Solutions

30 High-Frequency Placement Questions with Answers and Detailed Option-by-Option Explanations

Q1. OSI Model: Which OSI layer is responsible for transmitting raw bits over a physical medium?

A. Data Link | B. Network | C. Physical | D. Transport

Answer: C. Physical ✓

- **A. Data Link** ✗ — Handles frames, MAC addresses, and local-link communication.
- **B. Network** ✗ — Handles IP addressing and routing.
- **C. Physical** ✓ — Transmits raw bits/signals through cables, fiber, or wireless media.
- **D. Transport** ✗ — Handles end-to-end communication using TCP/UDP.

🧠 Physical = Bits

Q2. Data Link Layer: A network device forwards frames based on MAC addresses. Which OSI layer is primarily involved?

A. Physical | B. Data Link | C. Network | D. Transport

Answer: B. Data Link ✓

- **A. Physical** ✗ — Handles signals and bits.
- **B. Data Link** ✓ — Handles frames and MAC addresses.
- **C. Network** ✗ — Uses IP addresses and performs routing.
- **D. Transport** ✗ — Uses TCP/UDP.

🧠 **MAC** → **Data Link** → **Layer 2**

Q3. Network Layer: Which OSI layer is responsible for logical addressing and routing?

A. Transport | B. Session | C. Network | D. Data Link

Answer: C. Network ✓

- **A. Transport** ✗ — TCP/UDP and end-to-end delivery.
- **B. Session** ✗ — Manages sessions.
- **C. Network** ✓ — Handles logical addressing and routing.
- **D. Data Link** ✗ — Handles MAC addresses and frames.

🧠 **IP + Routing** → **Network Layer**

Q4. Transport Layer: Which protocols operate primarily at the Transport layer?

A. HTTP and FTP | B. IP and ICMP | C. TCP and UDP | D. Ethernet and Wi-Fi

Answer: C. TCP and UDP ✓

- **A. HTTP and FTP** ✗ — Application layer protocols.
- **B. IP and ICMP** ✗ — Primarily associated with the Network layer.
- **C. TCP and UDP** ✓ — Transport-layer protocols.
- **D. Ethernet and Wi-Fi** ✗ — Primarily Data Link/Physical technologies.

🧠 **Layer 4** → **TCP/UDP**

Q5. Application Layer: Which of the following is an Application-layer protocol?

A. HTTP | B. IP | C. TCP | D. Ethernet

Answer: A. HTTP ✓

- **A. HTTP** ✓ — Application-layer protocol used for web communication.
- **B. IP** ✗ — Network layer.
- **C. TCP** ✗ — Transport layer.
- **D. Ethernet** ✗ — Data Link/Physical technology.

Other Application-layer examples: DNS, HTTP, HTTPS, FTP, SMTP, SSH

Q6. Data Units: Which sequence is correct?

A. Physical → Frames, Data Link → Bits, Network → Packets
B. Physical → Bits, Data Link → Frames, Network → Packets
C. Physical → Packets, Data Link → Segments, Network → Bits
D. Physical → Data, Data Link → Packets, Network → Frames

Answer: B. Physical → Bits, Data Link → Frames, Network → Packets ✓

- **A.** ✗ — Frames belong to Data Link, not Physical.
- **B.** ✓ — Correct mapping.
- **C. & D.** ✗ — Incorrect mapping.

Memorize: Physical → Bits | Data Link → Frames | Network → Packets | Transport → Segments/Datagrams

Q7. Router: A company uses a device to forward packets between two different IP networks. Which device is being used?

A. Hub | B. Switch | C. Router | D. Repeater

Answer: C. Router ✓

- **A. Hub** ✗ — Broadcasts signals to ports.
- **B. Switch** ✗ — Primarily forwards frames using MAC addresses.
- **C. Router** ✓ — Routes packets between different IP networks.
- **D. Repeater** ✗ — Regenerates/repeats signals.

🧠 Different networks → Router

Q8. Switch: A switch receives an Ethernet frame and uses the destination MAC address to determine the appropriate outgoing port. Which layer is primarily involved?

A. Physical | B. Data Link | C. Network | D. Transport

Answer: B. Data Link ✓

- **A. Physical** ✗ — Signals/bits.
- **B. Data Link** ✓ — Ethernet switching primarily uses MAC addresses at Layer 2.
- **C. Network** ✗ — IP routing.
- **D. Transport** ✗ — TCP/UDP.

🧠 Switch → MAC → Layer 2 (Modern multilayer switches can route at L3, but standard basic MCQs = Layer 2).

Q9. Hub: A network device receives a signal and sends it out through all of its ports without making forwarding decisions based on MAC addresses. Which device is this?

A. Router | B. Switch | C. Hub | D. Gateway

Answer: C. Hub ✓

- **A. Router** ✗ — Routes packets using IP.
- **B. Switch** ✗ — Makes forwarding decisions using MAC addresses.
- **C. Hub** ✓ — Repeats incoming signals out multiple/all ports.
- **D. Gateway** ✗ — Connects networks/systems across different protocols.

🧠 Hub = Layer 1

Q10. TCP: A banking application requires data to arrive reliably and in the correct order. Which protocol is more appropriate?

A. UDP | B. TCP | C. ICMP | D. ARP

Answer: B. TCP ✓

Banking requires reliable and ordered delivery.

- **A. UDP** ✗ — Doesn't provide TCP-style reliable ordered delivery.
- **B. TCP** ✓ — Reliable, ordered, connection-oriented.
- **C. ICMP** ✗ — Used for network diagnostics/control messaging.
- **D. ARP** ✗ — Resolves IPv4 addresses to MAC addresses on local networks.

🧠 Reliable → TCP

Q11. UDP: An online gaming application prioritizes low latency and can tolerate occasional packet loss. Which protocol is generally more suitable?

A. TCP | B. UDP | C. FTP | D. SMTP

Answer: B. UDP ✓

Gaming often prioritizes low latency and can tolerate some packet loss.

- **A. TCP** ✗ — Has additional reliability mechanisms and overhead.
- **B. UDP** ✓ — Connectionless and lower overhead.
- **C. FTP** ✗ — File-transfer protocol.
- **D. SMTP** ✗ — Email protocol.

🧠 **Low latency / loss tolerance** → **UDP**

Q12. TCP Handshake: What is the correct order of the TCP three-way handshake?

A. ACK → SYN → SYN-ACK | B. SYN → ACK → SYN-ACK | C. SYN → SYN-ACK → ACK | D. SYN-ACK → SYN → ACK

Answer: C. SYN → SYN-ACK → ACK ✓

Step 1: Client → SYN → Server | Step 2: Client ← SYN-ACK ← Server | Step 3: Client → ACK → Server

🧠 **S-S-A : SYN → SYN-ACK → ACK**

Q13. TCP Characteristics: Which of the following is a characteristic of TCP?

A. Connectionless communication | B. No retransmission | C. Reliable and ordered delivery | D. No flow control

Answer: C. Reliable and ordered delivery ✓

- **A. Connectionless** ✗ — Characteristic of UDP.
- **B. No retransmission** ✗ — TCP can retransmit lost data.
- **C. Reliable and ordered** ✓ — Correct. (TCP also provides flow & congestion control).
- **D. No flow control** ✗ — TCP provides flow control.

Q14. UDP Characteristics: Which statement about UDP is correct?

A. It requires a three-way handshake | B. It guarantees delivery | C. It is connectionless and has low protocol overhead | D. It guarantees packet ordering

Answer: C. It is connectionless and has low protocol overhead ✓

- **A.** ✗ — TCP performs the three-way handshake.
- **B.** ✗ — UDP does not guarantee delivery.
- **C.** ✓ — Connectionless + low overhead.
- **D.** ✗ — UDP does not guarantee packet ordering.

🧠 **UDP = Fast/low overhead, but no TCP-style reliability**

Q15. IPv4: How many bits are present in an IPv4 address?

A. 16 | B. 32 | C. 64 | D. 128

Answer: B. 32 bits ✓

Example: 192.168.1.10 → Four octets × 8 bits: $4 \times 8 = 32$ bits. (128 bits is IPv6).

Q16. IPv6: How many bits are present in an IPv6 address?

A. 32 | B. 64 | C. 96 | D. 128

Answer: D. 128 bits ✓

🧠 **IPv4 → 32 bits | IPv6 → 128 bits**

Q17. Private IP: Which of the following is a private IPv4 address?

A. 8.8.8.8 | B. 172.20.10.5 | C. 1.1.1.1 | D. 142.250.1.1

Answer: B. 172.20.10.5 ✓

- A, C, D ✗ — Public IP addresses.
- B. ✓ — Falls within 172.16.0.0 – 172.31.255.255.

⚠ Important trap: Not every 172.x.x.x address is private. Only: 172.16.x.x → 172.31.x.x

Q18. Private IP Range: Which range is reserved for private IPv4 networks?

A. 11.0.0.0/8 | B. 10.0.0.0/8 | C. 100.0.0.0/8 | D. 200.0.0.0/8

Answer: B. 10.0.0.0/8 ✓

🧠 Remember the three private ranges: 10.x.x.x | 172.16.x.x – 172.31.x.x | 192.168.x.x

Q19. NAT: A home network has 20 devices using private IP addresses, but all devices access the Internet using one public IP address. Which technology commonly enables this?

A. DNS | B. DHCP | C. NAT | D. FTP

Answer: C. NAT ✓

- A. DNS ✗ — Name resolution | B. DHCP ✗ — Automatic IP config | D. FTP ✗ — File transfer
- C. NAT ✓ — Translates network addresses, allowing private hosts to share public IPv4 addresses.

🧠 Private IP ↔ Public IP translation → NAT

Q20. DNS: A user enters `www.example.com`. The computer needs to determine the corresponding IP address. Which service is primarily responsible?

A. DHCP | B. DNS | C. FTP | D. SMTP

Answer: B. DNS ✓

🧠 DNS = Domain Name → IP

Q21. DHCP: A newly connected laptop automatically receives an IP address, subnet mask, default gateway, and DNS server information. Which protocol is responsible?

A. DNS | B. DHCP | C. HTTP | D. ARP

Answer: B. DHCP ✓

- DHCP ✓ — Provides network configuration dynamically. (ARP resolves IPv4 address to MAC).

Q22. DHCP DORA: What is the correct order of the DHCP process?

A. Discover → Request → Offer → Acknowledgment | B. Offer → Discover → Request → Acknowledgment
C. Discover → Offer → Request → Acknowledgment | D. Request → Discover → Offer → Acknowledgment

Answer: C. Discover → Offer → Request → Acknowledgment ✓

🧠 DORA = Discover → Offer → Request → Acknowledgment

Q23. HTTP: Which port is commonly associated with HTTP?

A. 21 | B. 22 | C. 80 | D. 443

Answer: C. 80 ✓

21 = FTP control | 22 = SSH | 80 = HTTP | 443 = HTTPS

Q24. HTTPS: Which port is commonly associated with HTTPS?

A. 25 | B. 53 | C. 80 | D. 443

Answer: D. 443 ✓

🧠 HTTP → 80 | HTTPS → 443

Q25. FTP: Which protocol is primarily used for file transfer?

A. FTP | B. SMTP | C. DNS | D. DHCP

Answer: A. FTP ✓

🧠 FTP → File Transfer

Q26. SSH: A system administrator wants to securely log into a remote Linux server and execute commands. Which protocol should be used?

A. FTP | B. SSH | C. HTTP | D. SMTP

Answer: B. SSH ✓

🧠 SSH → Secure remote shell → Port 22

Q27. SMTP: Which protocol is primarily associated with sending/relaying email?

A. POP3 | B. IMAP | C. SMTP | D. DNS

Answer: C. SMTP ✓

• POP3 & IMAP = Email retrieval | SMTP = Sending/relaying email

🧠 SMTP → Send email

Q28. Protocol-Port: Which pair is correctly matched?

A. HTTP — 22 | B. SSH — 80 | C. DNS — 53 | D. HTTPS — 21

Answer: C. DNS — 53 ✓

• HTTP → 80 | SSH → 22 | DNS → 53 | HTTPS → 443 | FTP → 21 | SMTP → 25

Q29. 🔥 Scenario: A user opens a website using HTTPS. During communication, the connection should protect data from eavesdropping and tampering. Which technology provides the security for HTTPS?

A. DHCP | B. TLS | C. DNS | D. ARP

Answer: B. TLS ✓

• TLS ✓ — Provides cryptographic protection (confidentiality & integrity) for HTTPS communication.

🧠 HTTPS = HTTP secured with TLS

Q30. 🔥 Mixed Scenario: A company has the following requirements: (1) Devices within a LAN communicate using MAC addresses, (2) A router forwards packets between different networks, (3) TCP is used when reliable delivery is required, (4) DNS resolves domain names, (5) DHCP automatically assigns IP configuration. Which statement is correct?

- A. Switch → MAC, Router → IP, TCP → Reliable, DNS → Name resolution, DHCP → IP configuration
- B. Switch → IP, Router → MAC, UDP → Reliable, DNS → IP assignment, DHCP → Email
- C. Switch → TCP, Router → UDP, DNS → File transfer, DHCP → Encryption
- D. Switch → DNS, Router → DHCP, TCP → Name resolution, DNS → IP assignment

Answer: A. Switch → MAC, Router → IP, TCP → Reliable, DNS → Name resolution, DHCP → IP configuration ✓

Breakdown: Switch → MAC address | Router → IP address / routing | TCP → Reliable delivery | DNS → Domain name resolution | DHCP → Automatic IP configuration.

Why others are wrong: B has incorrect roles (Switch does not route IP, UDP is not reliable), C and D mix unrelated protocols and roles completely.